Linking Climate Drivers to U.S. Temperature Variability: Spectral, Wavelet, and ANN-Based Forecasting Approaches

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Abstract:

Climate variability in the United States arises from the interplay of atmospheric, oceanic, and solar drivers. This study examines long-term regional temperature anomalies across the Eastern, Southern, Northern, and Western United States from 1895–2024, using NOAA Climate at a Glance datasets with a 1910–2000 baseline. Monthly anomalies were analyzed alongside potential climate drivers, including atmospheric CO_2 concentrations, the Oceanic Niño Index (ENSO), and sunspot activity. The workflow included: (1) data cleaning and compilation, (2) correlation and trend analysis to assess relationships among variables, (3) spectral analysis to identify dominant periodicities, (4) wavelet analysis to examine how these relationships evolved over time, and (5) artificial neural network (ANN) to forecast 36 months of regional temperature anomalies, capturing general warming trends were reflecting region-specific variability. Results indicate consistent warming from 1960–2024 both nationally and regionally. Correlation analysis showed a strong positive relationship between CO_2 and temperature anomalies, while ENSO and solar irradiance exhibited no strong linear correlations. Wavelet analysis confirmed ENSO cycles of 2.5–5 years from 1960-2024; solar irradiance showing weaker periodic connections. Ultimately, this work establishes a baseline forecasting approach that can be extended in future studies to incorporate more complex predictive models.

Background Information:

- In recent years, significant changes in temperature have been noted, due to factors such as climate change, agriculture, industry, and economy[1].
- Changes in temperature has garnered attention due to susceptibility of human activities to extreme environmental challenges[1].
- Climate change is affected by a variety of elements: the ENSO phenomenon, solar radiation, sun spots, global warming, and human induced factors[1].
- ENSO is a major climate phenomenon involving interactions between the ocean and atmosphere[2].
  - Can lead to global climate anomalies, such as precipitation reductions, droughts, and forest erosion[2].
  - El Niño events are characterized by widespread warming of central & eastern tropical Pacific Ocean sea surface temperature of several degrees[2].
  - Periodically ENSO occurs every 2-7 years, with a cycle peak at 4 years[3].
- Climatic events can also be affected by the planet's outer atmosphere and sun activity[3].
  - Long-term fluctuations in the sun's magnetic field, especially from sunspot activity, impact climate variations[3].
  - There are significant correlations between sunspots and territorial climatic phenomena in 8-12 year frequency cycles[3].

Objective:

Investigate the regional and global variability of key climate drivers - ENSO, CO_2 concentration, and sun spots - and assess their impact on long-term temperature anomalies across 4 U.S. climate regions.

Method/Materials:

- **Data Acquisition**
  - Obtained monthly mean atmospheric CO_2 concentration from NOAA's Mauna Loa Observatory dataset.
  - Obtained ENSO data: Oceanic Nino Index → NOAA/PSL's metric for tracking ENSO. Each data value is the three month running mean.
  - Obtained Sunspot data: from R library, monthly mean relative sunspot numbers from 1749 to 1983. Collected at Swiss Federal Observatory, Zurich until 1960, then Tokyo Astronomical Observatory.
  - Monthly temperature anomaly data for different regions in the US. The different regions were broken as western, northern, southern, and eastern.
- **RStudio (v.4.2.1)**
  - Correlation and Detrending: Pearson and Spearman Correlation on Raw vs. Smoothed Data (Global and Regional Correlation Analysis Completed)
  - Spectral analysis (periodogram)
  - Wavelet Analysis
  - ANN forecast for temperature anomaly by region

Results:

- **Overall temperature anomaly**: as time progressed, the temperature anomaly is increasing. This suggests a global warming trend from 1960 to 2024.
- **Temperature Anomalies by US Regions** (Figure 1 and 2)
  - **East**: Slight warming trend over a period of time (1960-2024).
  - **North**: Steady increase in warming: shows multiple spikes and dips, indicating temperature variability for short periods of time.
  - **South**: Steady increase in warming: also has spikes and dips, not as prominent as north.
  - **West**: Slight warming trend over a period of time (1960-2024). However, after 2010, there was a sudden increase in warming after 2010.
- **CO_2 vs. temperature anomaly** is positively correlated, which can be seen more prominently after the noise has been removed in the data (Figure 3)
- **ENSO vs. temperature anomaly** doesn't appear to have a strong linear correlation (Figure 4).
  - No relationship between ENSO and long-term temperature anomalies.
- **Solar irradiance vs temperature anomaly** shows no strong linear correlation.
- **Spectral Analysis (Periodogram)**: ENSO and Solar Irradiance show cyclic patterns (Figure 5 and 6)
- **Wavelet Analysis Revealed:**
  - ENSO oscillates every 2.5 to 5 years consistently from 1960-2024 (Figure 5)/
  - Solar Irradiance doesn't reveal a consistent repeating cycle for this period of time (Figure 6).

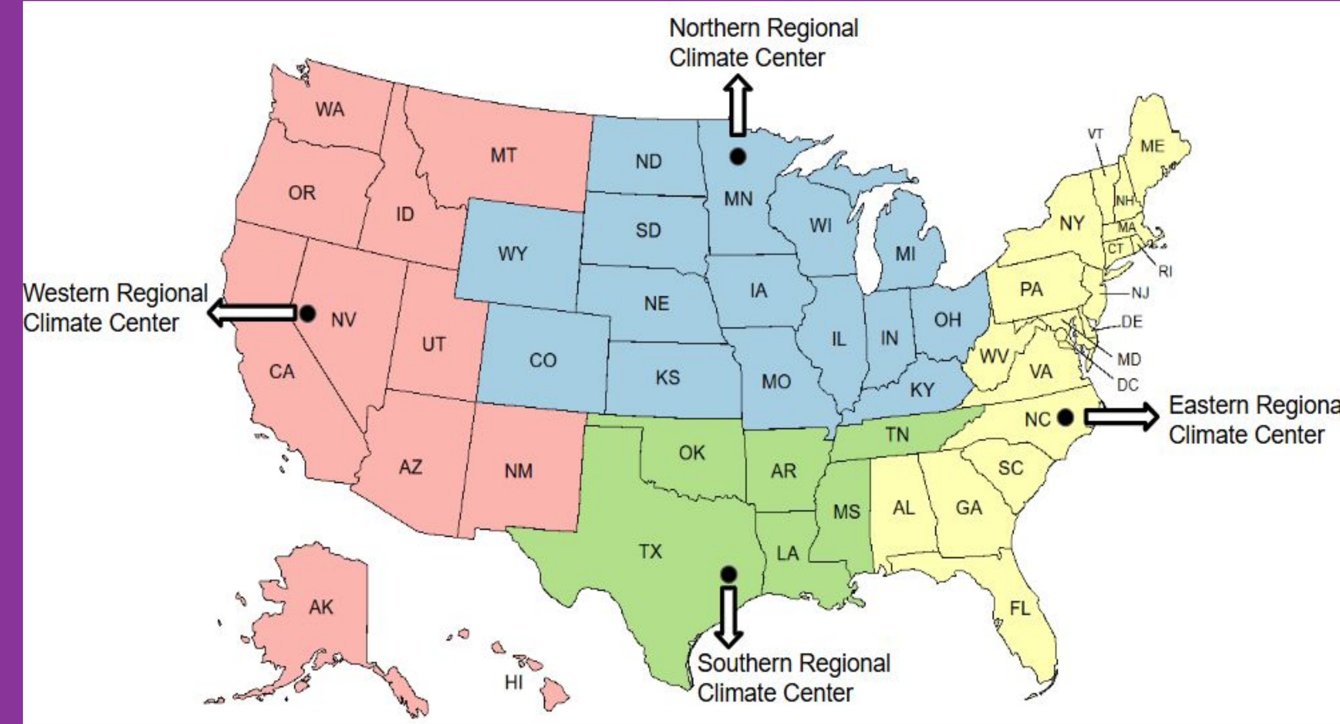

Figure 1: U.S. Map divided into 4 climate regions: north (blue), south (green), east (yellow), and west (red).

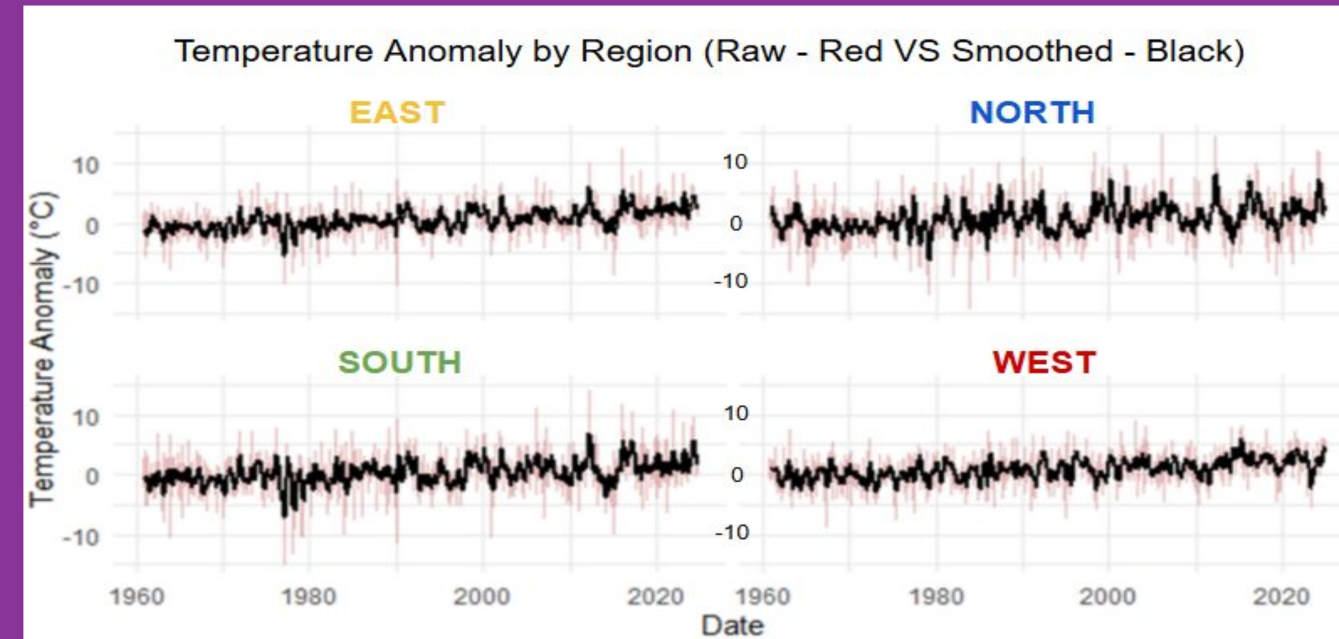

Figure 2: This figure depicts date vs. temperature anomaly by region. As time progressed, the temperature anomaly also increased, indicating that there is increased warming in all regions.

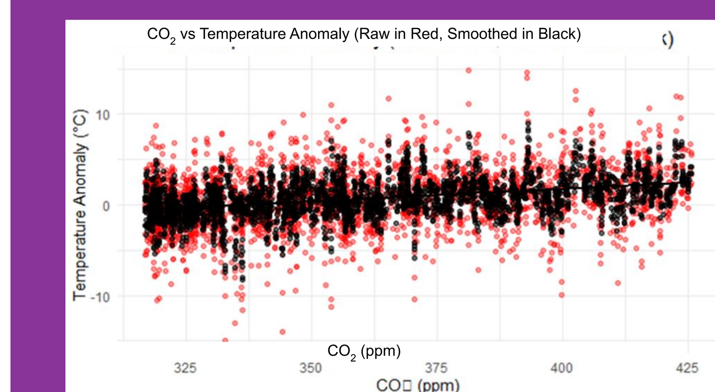

Figure 3: CO_2 vs temperature anomaly, where red represents the raw data and black represents the smoothed data. There is a positive correlation between CO_2 concentration and temperature anomaly, as the line of best fit trends upwards.

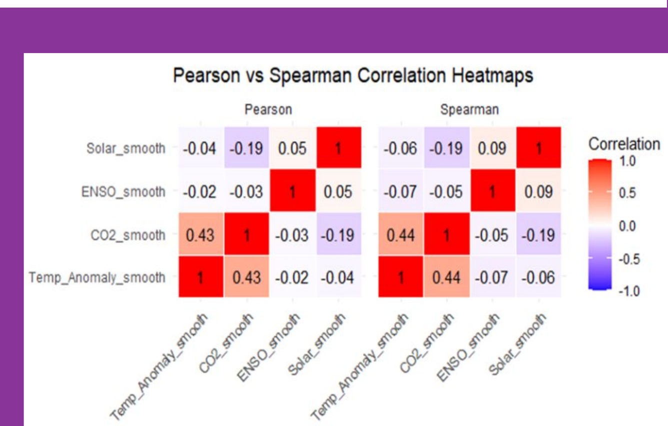

Figure 4: The matrix depicts the correlation level of different features. As it can be seen, CO_2 has a moderate, but positive, correlation with temperature anomalies. However, ENSO and solar factors don't seem to have a linear relationship with temperature anomalies.

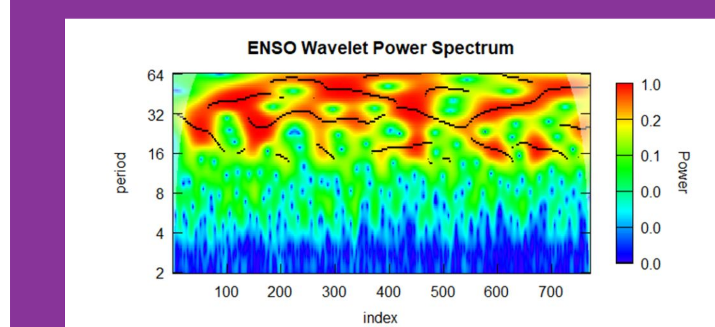

Figure 5: Wavelet power spectrum of the ENSO time series. The plot displays how ENSO signal power varies with period (y-axis, in years) and time (x-axis, in index units). Colors represent normalized wavelet power, with warmer colors (yellow–red) indicating stronger periodic signals and cooler colors (blue) indicating weaker signals. Strong power is observed in the 2.5–5 year period band, consistent with the typical ENSO cycle.

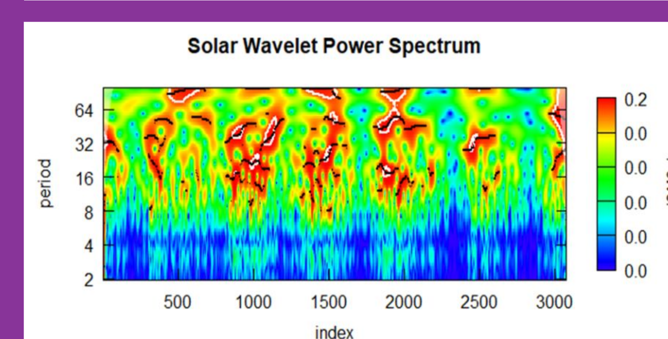

Figure 6: Wavelet power spectrum of solar time series. The plot displays how the solar signal power varies with time period. Color represent normalized wavelet power, with warmer colors (yellow-red) indicating stronger periodic signals and cool colors (blue) indicating weaker signals. Solar irradiance doesn't reveal a consistent repeating cycle for this period of time.

Discussion/Conclusion:

- This study investigated the effect of ENSO, CO_2 concentration, and sun spots on temperature anomaly.
  - Found that CO_2 concentration and temperature anomaly had a moderate, positive correlation.
  - ENSO and Solar irradiation didn't show a linear trend with temperature anomaly, but had cyclic patterns - this pattern was more pronounced for ENSO.
- Additionally, compared to ENSO and CO_2, solar irradiance had a smaller influence on temperature over this period of time (1960-2024).
  - Solar irradiance plays a role in Earth's long-term climate, and its impact on regional and global temperature anomalies appear minor in comparison to CO_2 emission and ENSO activity, which affects climate over the short-term.

Future Directions:

- Determine the weightage that each factor plays in temperature anomalies to make future predictions about climate.

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Credit Authorship Contribution Statement:

Adhithi: writing, Rstudio coding, maps, graphs, tables, correlation heatmap, poster layout and design, abstract, introduction, methods, results, discussion/ conclusions; Marsellos, A.E.: supervision

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